

PHY101
Assignment 2
Tutorial Date: October 31, 2025

1. Two identical blocks start from rest on different frictionless inclines that meet at the same bottom point. One incline is longer but less steep, the other shorter and steeper.
 - (a) Without detailed calculation, argue which block reaches the bottom first, if either.
 - (b) Verify by finding the time to reach the bottom as a function of incline angle θ and vertical drop h .
 - (c) Suppose the angle can be varied continuously while keeping h fixed. What angle minimizes the travel time?

2. A puck slides up a rough incline making an angle θ with the horizontal. The coefficient of kinetic friction is initially μ_1 , but after the puck has slid a distance d it drops suddenly to μ_2 ($\mu_2 < \mu_1$).
 - (a) Derive the total distance the puck travels before coming momentarily to rest.
 - (b) Does it slide back past the point where the coefficient changes? Give a physical explanation based on energy.
 - (c) For what combination of θ , μ_1 , μ_2 , and d will the puck never return to its starting point?

3. A bead of mass m slides without friction on a vertical circular wire loop of radius R . It is released from rest at an angle θ_0 above the lowest point.
 - (a) Find its speed and the wire tension at the bottom of the loop.
 - (b) Show that for small θ_0 the difference in tension between top and bottom depends only on m , g , and R , not θ_0 .
 - (c) If the wire were to break at the bottom, what would be the instantaneous angular momentum of the bead about the loop's center?

4. A particle moves along a line under a force $F(x) = -kx + \alpha|x|x$ with $k, \alpha > 0$.

- (a) Sketch $F(x)$ and locate all equilibrium points.
 - (b) For small oscillations about each equilibrium, find the approximate frequency.
 - (c) Describe qualitatively what happens as the amplitude increases enough for the particle to cross $x = 0$.
5. A spacecraft experiences a net force $F(r) = -k/r^2 + F_0$ directed inward, where F_0 is a small constant thrust.
- (a) Write down the effective potential and discuss whether bounded orbits can exist.
 - (b) Does the constant inward thrust affect angular-momentum conservation? Explain.
 - (c) For small F_0 , estimate how the radius of a circular orbit shifts compared with the case $F_0 = 0$.
6. A rocket of initial mass m_0 moves in deep space under a small external resistive force $F_{\text{ext}} = -\beta v$. The exhaust speed of the fuel relative to the rocket is u .
- (a) Derive the modified rocket equation including the drag term.
 - (b) Find the terminal velocity as the rocket mass decreases to m_f .
 - (c) Discuss qualitatively how the drag changes the efficiency compared to the ideal case without drag.
7. A small cart of mass m moves with speed v_0 toward an identical cart attached to a light spring of stiffness k whose other end is fixed to a platform that can move freely on a frictionless track.
- (a) Explain qualitatively what motions occur after the collision and why total momentum is still conserved.
 - (b) Using conservation laws, find the maximum compression of the spring.
 - (c) At that instant, what is the velocity of the platform?
8. At a certain instant, a particle under an inverse-square central force has position r and velocity making an angle ϕ with the radius vector.

- (a) Express its instantaneous angular momentum L and total energy E in terms of r , v , and ϕ .
 - (b) Without integrating, determine how the sign of E relates to the type of trajectory (ellipse, parabola, or hyperbola).
 - (c) If a brief external torque changes only the direction of v but not its magnitude, what happens to L and E ?
- 9.** A particle moves in the xy -plane under a force $\mathbf{F} = kx \hat{\mathbf{y}}$.
- (a) Calculate the torque about the origin and show that $dL_z/dt \neq 0$.
 - (b) Even though there is a nonzero torque, can the particle ever move in a circle about the origin? Explain.
 - (c) Describe qualitatively how the angular momentum evolves with time.
- 10.** A charged particle of mass m and charge q moves in a uniform magnetic field $\mathbf{B} = B \hat{\mathbf{z}}$.
- (a) Solve the equations of motion to find its trajectory.
 - (b) Compute its angular momentum about the origin and discuss whether it is conserved.
 - (c) Is the mechanical energy constant? Explain briefly in terms of the magnetic force.

End of Assignment 2.